

A Staged Multi-Agent Validator for Scientific Claim Verification

Multi-Agent Validator System

Abstract

We study an inference-time validator that checks scientific claims against original citations (D0) and independently retrieved abstracts (D1). The system stages a sealed evidence reader, a judge, a citation auditor, and a reconciler, then maps four-way internal labels onto SciFact’s SUPPORT / REFUTE / NEI space. The preregistered primary comparison is false endorsement of Validator V versus same-evidence baseline B2. This paper reports a SciFact-native evaluation: we do not add human four-way gold or a human study. Those remain future work. If V does not reduce false endorsement relative to B2, we report that as a negative result on the same tables.

1 Introduction

Large language models are increasingly used to write and check scientific claims. A system that endorses a claim the evidence does not support is more harmful than one that abstains. We ask whether a staged, evidence-first validator reduces that false-endorsement risk compared with a same-evidence baseline that already reads the original citations.

Research questions retained in this paper:

- RQ1 (primary): Does V reduce false endorsement versus B2 on SciFact?
- RQ2 (descriptive only): How does V split predicted NEI into unaddressed versus underdetermined? SciFact gold cannot score this.
- RQ4 (secondary): How do local and API model families compare on cost and the same metrics?

RQ3 (human study) is deferred.

2 Related work

Fact verification corpora such as FEVER and SciFact supply claim–evidence pairs with a three-way label space. LLM-as-judge work shows that a single model can assign those labels, but it also hallucinates support. Citation auditing and selective prediction (risk-coverage) are the closest mechanisms to our fail-closed auditor and confidence thresholds. We do not claim a reproduction of trained verifiers such as KGAT or ARISTO; named baselines here are prompt adaptations.

3 Method

The validator is a staged pipeline, not a single prompt.

Reader. A sealed, claim-blind reader sees a neutral question and shuffled passages. It must not see the asserted answer or gold rationales.

Judge. A claim-visible judge assigns a four-way label (supported, contradicted, unaddressed, underdetermined) to one evidence scope.

Citation auditor. D0 original citations are audited for adequacy. Structural defects and parse failures are fail-closed.

Reconciler. D0 and D1 judgments are combined. Agreement copies the shared four-way label. Conflict or a failed D0 audit is underdetermined, then mapped to NEI.

Label mapping. supported→SUPPORT, contradicted→REFUTE, unaddressed and underdetermined→NEI. The mapping is the only bridge onto SciFact gold.

4 Experimental setup

Data. SciFact abstracts, locked corpus hash in `configs/corpus/scifact.yaml`. Splits: calibration 100, development 709, held-out local eval 300 (public SciFact dev), official test 300 unlabeled. The n=300 development sample is a seeded draw (`data/manifests/dev300_seed42.json`); the held-out 300 is unused until one frozen run.

Design. Evidence access (D0 vs D1) × reading policy (claim-visible vs neutral evidence-first). Cells: B1, B2, B3, V-D1, plus full V, prior-knowledge B0, and oracle O. Matched-budget B2@*k* uses a token-matched self-consistency vote.

Statistics. Paired cluster bootstrap, 2000 resamples, 95% percentile interval. Families are connected components over shared `cited_doc_ids`. Holm adjustment is applied to secondary metrics only. The primary metric is false endorsement on the gold non-support denominator.

What we do not claim. No human annotation, no four-way gold, no sentence-level official evidence score, no human study.

5 Results

On the frozen development 300 (gpt-6-luna), B2 scores micro-F1 0.747 and false endorsement 0.011 [0.000, 0.031]. V never false-endorses on the 268 paired execution-ok claims (FE 0.000), but SUPPORT recall falls from 0.530 to 0.383 and 32 V rows fail closed. The paired FE difference (V − B2) is −0.012 [−0.031, 0.000] and does not exclude zero ($p = 0.46$). Following the preregistered decision rule, we report a null primary result: the staged validator does not significantly reduce false endorsement relative to B2.

The claim-visible D0 judge (B1) is stronger on F1 (0.840) with FE 0.049. Prior-knowledge B0 has FE 0.448. The gold-rationale oracle reaches micro-F1 0.857. V-D1 (D1 arm only) has micro-F1 0.743 and FE 0.019. Of V’s 225 predicted NEI rows, 60 are unaddressed and 165 are underdetermined; SciFact gold cannot score that split.

Held-out 300 (one frozen run, 219 families, 262 paired claims) reproduces the same pattern: B2 micro-F1 0.760 and FE 0.013 [0.000, 0.033]; V FE 0.000 with micro-F1 0.618 and 38 fail-closed rows. The paired difference is -0.013 $[-0.033, 0.000]$ and still includes zero ($p = 0.44$).

6 Analysis

If both methods rarely predict SUPPORT, RQ1 cannot discriminate endorsement errors and the interesting contrast is precision versus coverage (risk-coverage). V’s reason codes (D0/D1 conflict, fail-closed, citation defects) are the descriptive RQ2 table.

7 Limitations and future work

SciFact NEI is not four-way gold. We do not recover unaddressed versus underdetermined as scored classes. Sentence-level evidence selection is unscored. The human study and agentic-annotation protocol remain future work. API model access and cost limit the number of families we can compare.

8 Ethics

The task is interpreting published abstracts, not giving medical advice. Outputs are labels and rationales over a frozen corpus. The API key is never logged or committed.

A Prompts

Versioned templates live under `prompts/`. The run sidecar records the prompt-bundle hash.

B Freeze file

The held-out run is allowed only when `--frozen-final` matches `docs/freeze.json` (config, prompt, corpus, and model hashes).